

COIN CHANGE VISUALIZER

A Project Report

Submitted by

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In partial fulfilment for the requirements of the Coding Skills Project

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING



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CERTIFICATE

This is to certify that the project work entitled “**Coin Change Visualizer**” is a bonafide record of project work carried out by the following students:

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The students have carried out this project work under my supervision during the period **January 2026 to April 2026**.

It is further certified that, to the best of my knowledge, this project has not previously formed the basis for the award of any degree or any other similar title to this or any other candidate.

This is also to certify that the project work represents the teamwork of the candidates.

Station: Mangalagiri

Date: 22-04-2026

SRM University-AP
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ABSTRACT

The *Coin Change Visualizer* is a web-based system developed to demonstrate and compare two important algorithmic approaches used to solve the Coin Change Problem: the Greedy Algorithm and Dynamic Programming (DP). The system aims to improve understanding of algorithm behavior through interactive and visual representation.

The application consists of a user-friendly interface where users can input a target amount and coin denominations. The system then allows users to choose between Greedy and Dynamic Programming approaches to visualize how the solution is formed step-by-step.

Users can observe how the Greedy algorithm selects coins based on immediate best choices, which may not always produce optimal results. In contrast, Dynamic Programming ensures the optimal solution by solving subproblems and building up the final answer.

The main objective of this project is to simplify the learning of algorithms by providing a visual and interactive platform. It reduces the complexity of understanding theoretical concepts and allows users to experiment with different inputs. The system is efficient, responsive, and accessible through any modern web browser without requiring additional setup.

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1. INTRODUCTION

Understanding algorithms through theory alone can be difficult for students. The Coin Change Problem is a classic problem used to demonstrate algorithm design techniques such as Greedy and Dynamic Programming.

This project presents a *Coin Change Visualizer*, which provides an interactive platform to understand how these algorithms work. The system allows users to input values and observe step-by-step execution of algorithms.

The platform improves learning by showing how solutions are formed and highlighting differences between optimal and non-optimal approaches.

1.1 Access Specification

There is only one type of user in this system:

User:

- Inputs target amount and coin denominations
 - Chooses algorithm (Greedy or DP)
 - Views step-by-step visualization
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2. REQUIREMENT ANALYSIS

2.1 Functional Requirements

- User can enter target amount
 - User can input coin denominations
 - User can select Greedy algorithm
 - User can select Dynamic Programming algorithm
 - System displays step-by-step visualization
 - System shows final result (coins used and count)
-

2.2 Non-Functional Requirements

2.2.1 Performance/Time

The system provides fast execution and smooth visualization. Algorithms run efficiently even for moderate input sizes.

2.2.2 Usability

The interface is simple and user-friendly. Users can easily understand and interact with the system without prior experience.

2.2.3 Security

Since the system does not store personal data, security requirements are minimal. Input validation is implemented to prevent errors.

2.2.4 Reliability and Availability

The system runs reliably in modern browsers and is available anytime without requiring installation.

3. CONCEPTUAL MODELS

3.1 Algorithm Flow Representation

The system represents algorithm execution step-by-step, showing how coins are selected and how the solution is built.

3.2 Logical Structure

The system consists of:

- Input module
- Algorithm processing module
- Visualization module

4. SYSTEM DESIGN

4.1 Design Goals

- Simplicity: Easy to understand interface
 - Interactivity: Real-time visualization
- Efficiency: Fast execution of algorithms
- Responsiveness: Works across devices

4.2 Design Development

The system follows a **client-side architecture** using:

- HTML for structure
- CSS for design
- JavaScript for logic and visualization

Data flows from user input → algorithm processing → visualization output.

4.3 Detailed Design Understanding

4.3.1 Algorithm Logic

Greedy Approach:

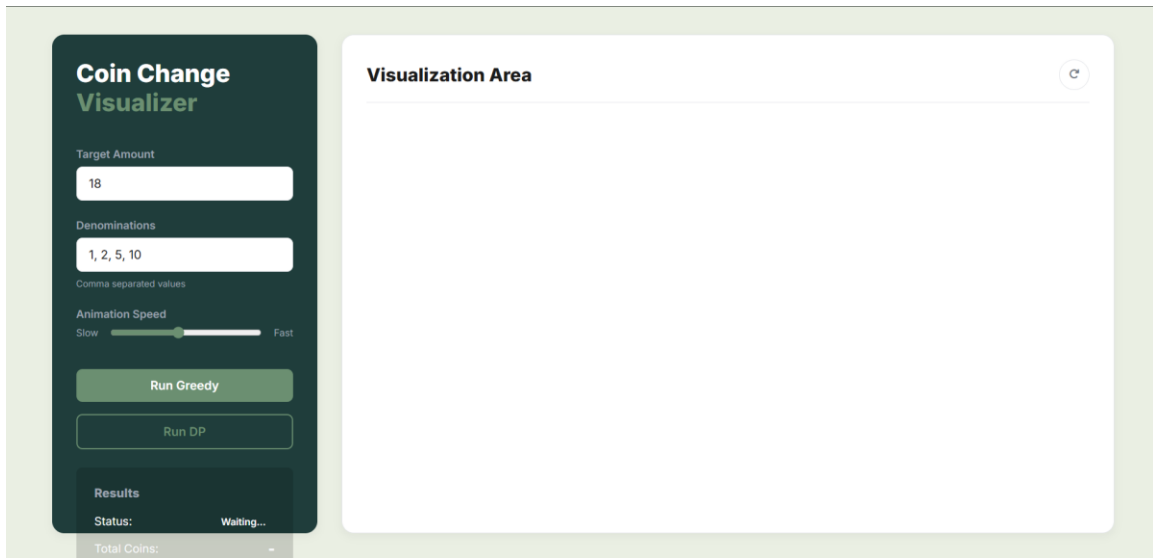
- Select largest coin first
- Repeat until amount is reached

Dynamic Programming:

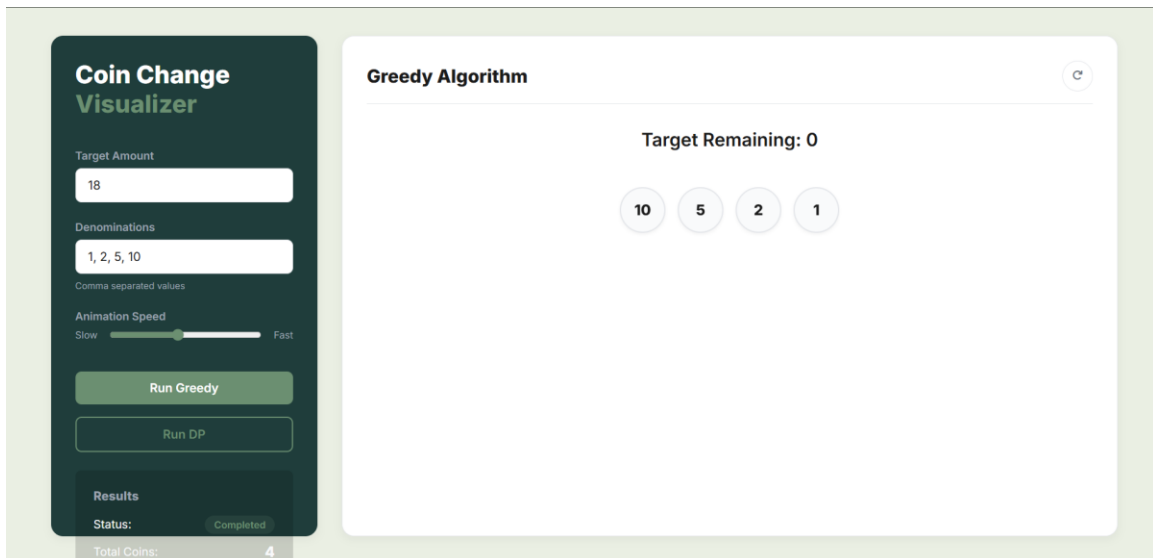
- Solve smaller subproblems
 - Store results in an array
 - Build optimal solution
-

4.3.2 Screenshots

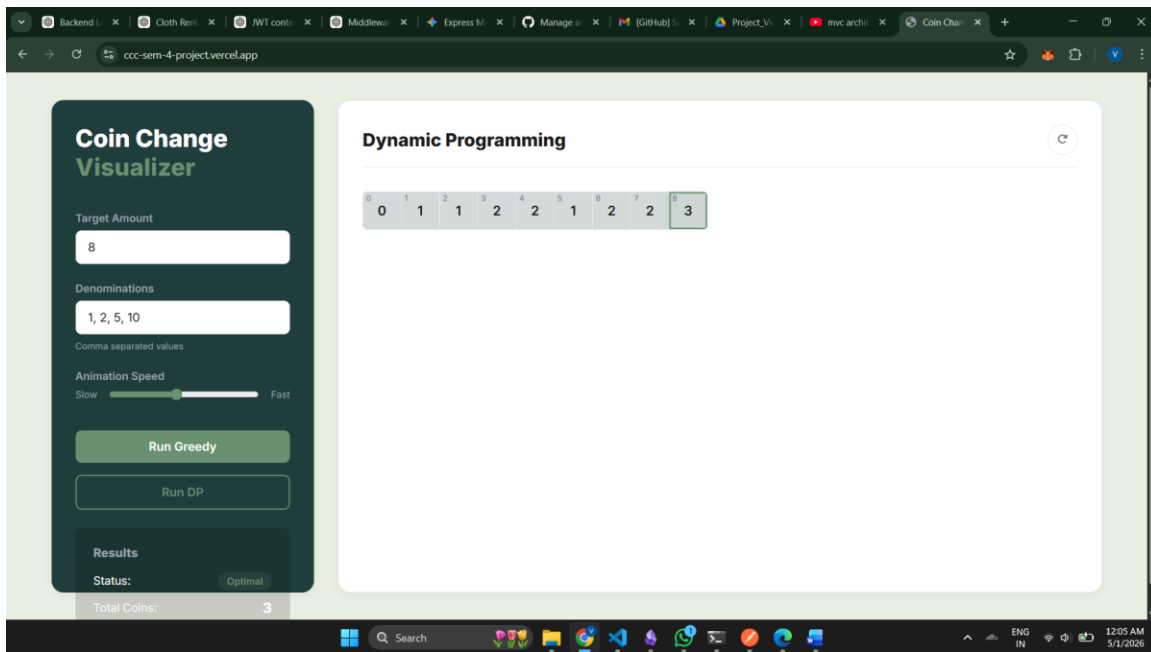
- Input interface



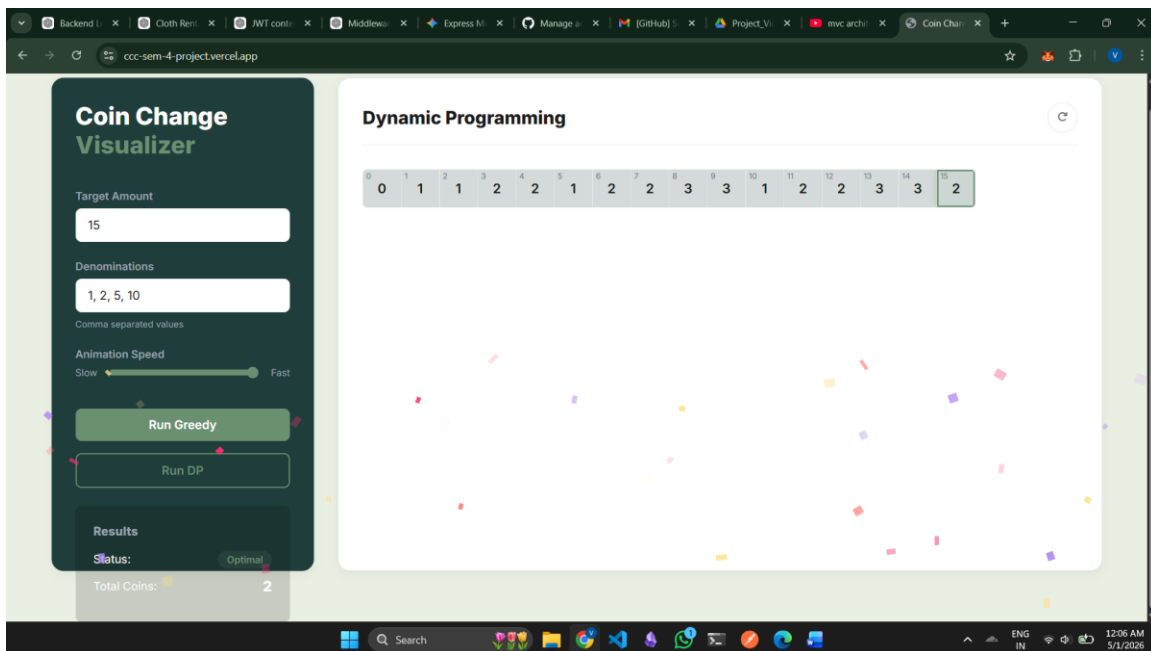
- Greedy visualization



- Dynamic Programming visualization



- Final output display



5. CONCLUSION

The Coin Change Visualizer provides an effective way to understand algorithm behavior through visualization. It

highlights the difference between Greedy and Dynamic Programming approaches clearly.

The project simplifies complex concepts and makes learning interactive and engaging. It serves as a useful educational tool for students studying algorithms.

6. REFERENCES

- Introduction to Algorithms – Cormen
- GeeksforGeeks (Coin Change Problem)
- MDN Web Docs (HTML, CSS, JavaScript)
- W3Schools Web Development Tutorials